

Advanced Communication Networks

شبکه‌های ارتباطی پیشرفته

Module 17

3GPP Standardization for Advanced Communication Networks

استانداردسازی 3GPP برای شبکه‌های ارتباطی پیشرفته



Lecture Notes

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1 What is 3GPP?

The **3rd Generation Partnership Project (3GPP)** is a global collaboration of telecommunications standards organizations that develops technical specifications and reports for mobile communication systems. Founded in December 1998, 3GPP unites seven regional Organizational Partners (OPs) to produce globally applicable standards covering radio access, core network architecture, and service capabilities.

Organizational Partners

Partner	Region	Full Name
ETSI	Europe	European Telecommunications Standards Institute
ARIB	Japan	Association of Radio Industries and Businesses
ATIS	North America	Alliance for Telecommunications Industry Solutions
CCSA	China	China Communications Standards Association
TTA	South Korea	Telecommunications Technology Association
TTC	Japan	Telecommunication Technology Committee
TSDSI	India	Telecommunications Standards Development Society, India

■ Key Characteristics

- **Not a legal entity** — 3GPP is a partnership project governed by its OPs
- **Market Representation Partners (MRPs)** provide industry input (e.g., GSMA, 5GAA, NGMN)
- **Individual Members** are companies participating through their regional OP
- Specifications are **transposed** into regional standards by each OP (e.g., ETSI publishes them as EN/ES/TS)
- Covers **2G (GSM/GPRS/EDGE), 3G (UMTS), 4G (LTE/LTE-A), and 5G (NR)** systems

2 3GPP Organizational Structure

■ Organizational Structure Diagram

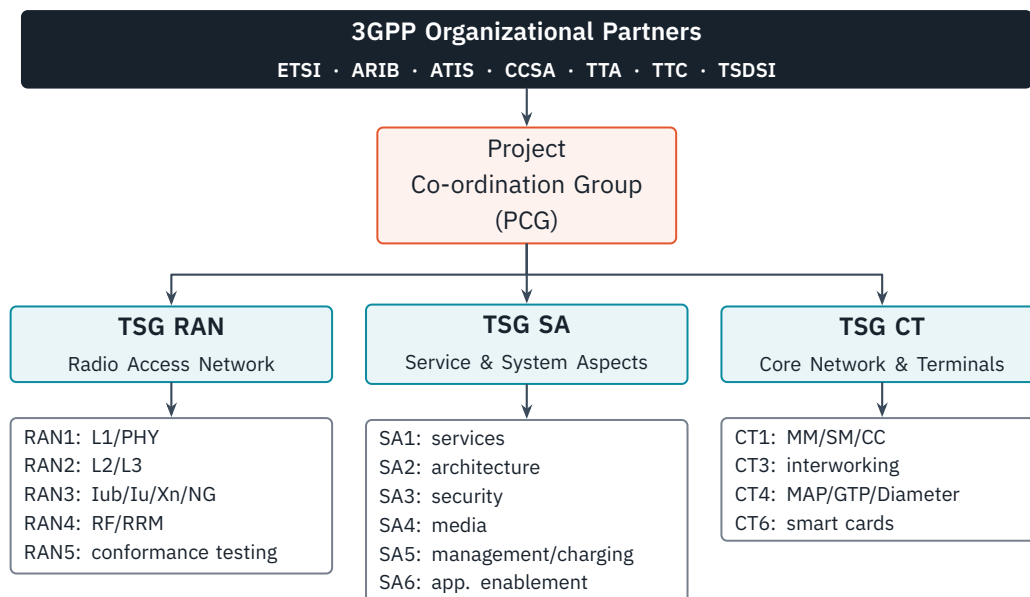


Figure 17.1 3GPP organisational structure

■ Technical Specification Groups (TSGs)

■ TSG RAN – Radio Access Network

Working Group	Responsibility
RAN1	Physical layer (PHY) specifications
RAN2	Layer 2 (MAC, RLC, PDCP, SDAP) and Layer 3 (RRC)
RAN3	Network architecture interfaces (Xn, NG, F1, E1)
RAN4	Radio performance, RF parameters, RRM requirements
RAN5	UE conformance testing

■ TSG SA – Service and System Aspects

Working Group	Responsibility
SA1	Service requirements and features
SA2	System architecture and network functions
SA3	Security and privacy
SA4	Media codecs, streaming, MBMS
SA5	Management, orchestration, and charging
SA6	Application layer enablement (SEAL, CAPIF, EDGEAPP)

■ TSG CT – Core Network and Terminals

Working Group	Responsibility
CT1	NAS protocols (MM, SM, call control)
CT3	Interworking with external networks
CT4	Control plane protocols (GTP-C, Diameter, HTTP/2-based SBI)
CT6	Smart card application aspects (USIM, ISIM)

3 Specification Types

3.1 Types of Documents

Type	Name	Nature	Purpose
TS	Technical Specification	Normative	Defines mandatory requirements, procedures, protocols
TR	Technical Report	Informational	Documents studies, feasibility analyses, best practices

- **TS documents** contain “shall” requirements — they define what implementations **MUST** do
- **TR documents** contain “should/may” language — they inform decisions but are not binding
- TRs often precede TSs (a study item produces a TR, which leads to a work item producing/updating TSs)

3.2 Numbering System

The specification number format is: **xx.yyy** (or xx.yyy for series ≤ 13) and **xx.yyy** for all series.

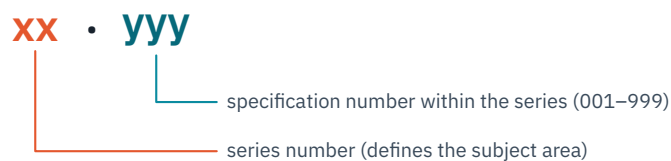


Figure 17.2 Specification numbering: series and number

Example: 38.331

- **38** = 5G NR radio technology series
- **331** = RRC protocol specification

■ Series 00–13: Legacy GSM/GPRS Series (originally numbered without leading digits)

These map to the older GSM specification numbering and are maintained for backward compatibility.

■ Series 21–38: UMTS and Beyond

Series	Subject Area
21	Requirements
22	Service aspects (“stage 1”)
23	Technical realization (“stage 2” — architecture)
24	Signalling protocols (“stage 3” — UE to network)
25	UTRAN radio technology (3G)
26	Codecs
27	Data (AT commands, etc.)
28	Signalling protocols (“stage 3” — SS, OAM)
29	Core network protocols (stage 3 — network-to-network)
30	Programme management (Rel-4 old numbering)
31	UICC / USIM / ISIM
32	OAM&P (management and charging)
33	Security
34	UE test specifications
35	Security algorithms
36	LTE (Evolved UTRA) radio technology
37	Radio measurement and protocol conformance (multi-RAT)
38	5G NR radio technology

3.3 Stage Methodology (ITU-T approach)

Stage	Description	Typical Series
Stage 1	Service description from user’s perspective	22.xxx
Stage 2	Logical analysis / architecture	23.xxx
Stage 3	Protocol implementation / detailed procedures	24.xxx, 29.xxx

4 Release System

4.1 Release Lifecycle



Figure 17.3 Life cycle of a 3GPP release

Phase Details:

1. **Study Phase** — Feasibility and technical investigations; output is a TR
2. **Work Item Approval** — PCG/TSG approves; specification drafting begins
3. **Functional Freeze** — All features for the release are complete; no new functionality added
4. **ASN.1 / Protocol Freeze** — Protocol encoding (ASN.1, JSON, etc.) is frozen; only essential corrections
5. **Maintenance** — Only critical bug-fix Change Requests (CRs) accepted

4.2 Key Releases

Release	Year	Generation	Key Features
R99	2000	3G (UMTS)	WCDMA air interface, CS/PS core, 384 kbps DL
Rel-4	2001	3G	TD-SCDMA, MSC server/MGW split
Rel-5	2002	3G/3.5G	HSDPA (14.4 Mbps DL), IMS introduction
Rel-6	2005	3.5G	HSUPA (5.76 Mbps UL), MBMS, WLAN interworking
Rel-7	2007	3.5G+	HSPA+ (MIMO, 64QAM), VoIP over HSPA
Rel-8	2009	4G (LTE)	OFDMA/SC-FDMA, flat architecture (eNB/EPC), 100 Mbps DL
Rel-9	2010	4G	eMBMS, dual-layer beamforming, HeNB
Rel-10	2011	4G (LTE-A)	Carrier aggregation, 8×8 MIMO, relay nodes, 1 Gbps DL
Rel-11	2013	4G+	CoMP, enhanced ICIC, carrier aggregation enhancements
Rel-12	2015	4G+	D2D (ProSe), MTC/M2M, dual connectivity, 256QAM DL
Rel-13	2016	4G+	LAA, LTE-U, NB-IoT, eMTC (Cat-M1)
Rel-14	2017	4G+/pre-5G	V2X (LTE), eLAA, further IoT enhancements
Rel-15	2019	5G Phase 1	NR NSA/SA, eMBB, mmWave, 5GC (SBA), network slicing
Rel-16	2020	5G Phase 2	URLLC enhancements, NR V2X, IAB, NR-U, IIoT, NPN
Rel-17	2022	5G	NTN (satellite), RedCap, NR sidelink enh., XR, multicast
Rel-18	2024	5G-Advanced	AI/ML for NR, duplex evolution, NR-light enh., XR enh.
Rel-19	2025-26	5G-Advanced+	AI-native air interface studies, ambient IoT, further NTN

5 How to Navigate 3GPP Specifications

5.1 Spec-Numbering Convention

Understanding the number tells you immediately what a spec covers:

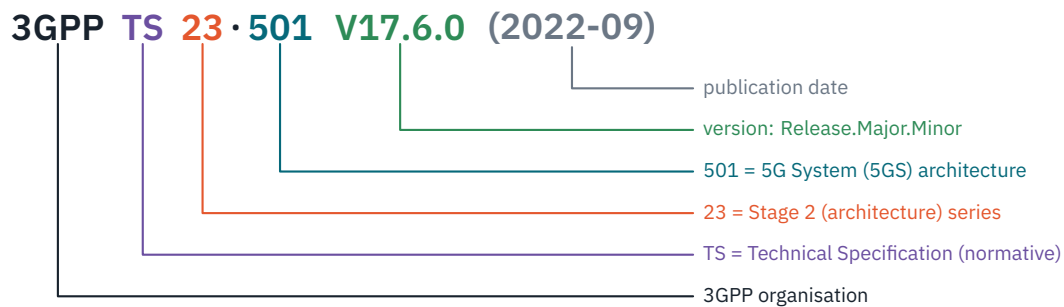


Figure 17.4 Reading a specification reference

Version numbering: The first digit of the version maps to the Release:

- Version 3.x.x → Release 99
- Version 4.x.x → Release 4
- ...
- Version 15.x.x → Release 15
- Version 17.x.x → Release 17

5.2 How to Find Specifications

Resource	URL	Use
Spec Finder	https://www.3gpp.org/specifications/specification-numbering	Search by number
Specification series	https://www.3gpp.org/specifications	Browse by series
Portal (latest + archive)	https://portal.3gpp.org/	Advanced search, CR tracking
FTP server	https://www.3gpp.org/ftp/Specs/	Direct download of .zip files
Work Plan	https://www.3gpp.org/specifications/work-plan	Track ongoing items

Practical Search Workflow:

- Go to **portal.3gpp.org** → “Specifications” tab
- Enter spec number (e.g., “23.501”) or keyword

3. Filter by Release and version
4. Download the latest approved version (.docx or .pdf)

5.3 Reading a 3GPP Specification

Every 3GPP spec follows a standard structure:

Section	Content
1. Scope	What the spec covers and does NOT cover
2. References	Normative and informative references to other specs
3. Definitions and abbreviations	Precise term definitions; abbreviations list
4. General / Overview	High-level system description
5+. Technical content	Procedures, architecture, message flows
Annexes	Additional information, ASN.1, examples

Tips for Reading:

- Start with **Scope** to confirm you have the right document
- Use **References** to trace dependencies (specs reference each other heavily)
- The **Definitions** section resolves ambiguity — always consult it
- Look for “**shall**” (mandatory), “**should**” (recommended), “**may**” (optional)
- Use Ctrl+F for specific procedure names or information elements

5.4 Key Specifications Every Telecom Engineer Should Know

Spec	Title	Why It Matters
TS 23.501	System architecture for the 5G System (5GS)	Defines 5GC architecture, network functions, reference points
TS 23.502	Procedures for the 5G System	Registration, PDU session, handover procedures
TS 24.501	NAS protocol for 5GS (UE ↔ AMF)	5G NAS message definitions, state machines
TS 38.300	NR and NG-RAN overall description	NR system overview, protocol stack, channels
TS 38.331	NR RRC protocol specification	RRC procedures, measurements, reconfiguration
TS 38.211	NR physical channels and modulation	OFDM parameters, resource grid, reference signals
TS 38.213	NR physical layer control procedures	DCI formats, HARQ, power control
TS 38.214	NR physical layer data procedures	Transport block size, CQI/MCS tables, scheduling
TS 33.501	Security architecture for 5GS	5G security framework, key hierarchy, algorithms
TS 28.541	5G NRM (Network Resource Model)	Network management and orchestration model

6 Study Items vs Work Items

6.1 Overview

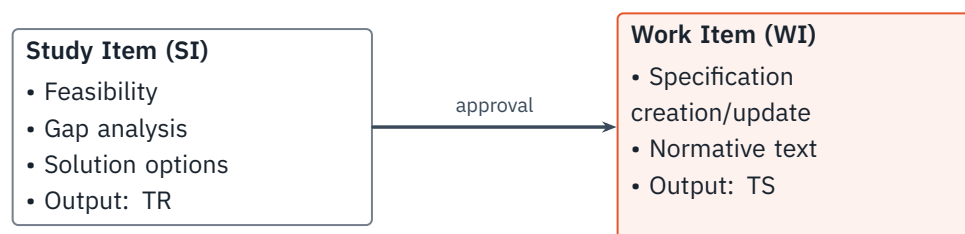


Figure 17.5 Study item versus work item

6.2 Study Item (SI)

- **Purpose:** Investigate feasibility of a new feature or technology
- **Output:** Technical Report (TR) documenting findings, options, conclusions
- **Duration:** Typically 1–4 TSG meetings (9–18 months)
- **Process:**

1. Company submits SI Description (SID) to TSG
2. TSG approves the study
3. Working group conducts the study (contributions from member companies)
4. TR is produced with conclusions and recommendations
5. If feasible → WI proposal submitted

6.3 Work Item (WI)

- **Purpose:** Create or modify normative specifications
- **Output:** New or updated Technical Specifications (TS)
- **Duration:** Tied to the target Release timeline
- **Process:**
 1. Company submits Work Item Description (WID) based on SI conclusions
 2. TSG approves the WI (assigns to working group, sets milestones)
 3. Working group drafts Change Requests (CRs) to existing specs or new specs
 4. CRs are reviewed, revised, and approved at WG/TSG level
 5. Approved CRs are incorporated into the spec version for that Release

6.4 SI → WI Flow Example

Example: NR Sidelink Relay (Release 17)

SI Phase (2019-2020):

- └─ TR 38.836: "Study on NR sidelink relay"
 - └─ Conclusions: Relay architectures A and B feasible
 - └─ Recommended for WI

WI Phase (2020-2022):

- └─ WID approved at TSG RAN#90
 - └─ Updates to: TS 38.300, TS 38.331, TS 38.321, TS 24.587
 - └─ Feature complete at Rel-17 functional freeze

7 Practical Guide: Referencing 3GPP Specs in Academic Work

7.1 Standard Citation Format

3GPP, "Title of Specification," 3GPP TS/TR xx.yyy, Version Vx.y.z, Month Year.

Example (IEEE style):

[1] 3GPP, "System architecture for the 5G System (5GS); Stage 2,"
3GPP TS 23.501, Version 17.6.0, Sep. 2022.

Example (APA style):

3rd Generation Partnership Project. (2022). System architecture for the 5G System (5GS); Stage 2 (3GPP TS 23.501 V17.6.0). <https://www.3gpp.org/>

7.2 Best Practices

Do	Don't
✓ Cite specific version (Vx.y.z)	✗ Cite without version (specs change every quarter)
✓ Include the Release number	✗ Assume "latest" is stable
✓ Reference the exact section/clause	✗ Cite entire 500-page spec without clause
✓ Use official 3GPP document titles	✗ Paraphrase titles
✓ Include access URL/date for online refs	✗ Link to unofficial mirrors

7.3 Where to Download for Citation

- **Official archive:** <https://portal.3gpp.org/desktopmodules/Specifications/SpecificationDetails.aspx?specificationId=xxxx>
- **Direct spec URL pattern:** [https://www.3gpp.org/ftp/Specs/archive/\[series\]/\[spec_number\]/](https://www.3gpp.org/ftp/Specs/archive/[series]/[spec_number]/)
- Always cite the **approved** version (not drafts or working documents)

7.4 Citing Change Requests (CRs) and Contributions

For referencing specific proposals or changes:

3GPP TSG RAN WG1, "Title," Tdoc R1-21xxxxx, Meeting #10x, Month Year.

8 Important Specification Series Table

Series	Area	Description	Example Specs
21	Requirements	Overall system requirements	21.915 (Rel-15 summary)
22	Services (Stage 1)	Service descriptions, use cases	22.261 (5G service requirements)
23	Architecture (Stage 2)	System architecture, functional descriptions	23.501 (5GS arch), 23.502 (5GS procedures)
24	UE-Network Signalling (Stage 3)	NAS protocols, SIP, AT commands	24.501 (5G NAS), 24.229 (IMS SIP)
25	UTRAN (3G Radio)	WCDMA physical layer, RLC, RRC	25.331 (UTRAN RRC)
26	Codecs	Speech, audio, video codecs	26.171 (AMR-WB), 26.346 (MBMS)
27	Data	Terminal adaptation, AT commands	27.007 (AT commands)
28	Signalling (Stage 3 - SS)	Supplementary services, OAM	28.541 (5G NRM)
29	CN Protocols (Stage 3)	Core network protocols (N-N)	29.500 (5GC HTTP APIs), 29.244 (PFCP)
30	Programme management	(Legacy - Rel-4 era)	—
31	UICC/USIM/ISIM	SIM card specifications	31.102 (USIM)
32	OAM & Charging	Management, performance, charging	32.255 (5G charging)
33	Security	Security architecture, algorithms	33.501 (5G security)
34	UE Testing	Test specifications, conformance	34.229 (SIP conformance)
35	Algorithms	Cryptographic algorithm specs	35.231 (TUAK algorithm)
36	LTE (E-UTRA)	LTE radio technology	36.331 (LTE RRC), 36.211 (LTE PHY)
37	Multi-RAT	Radio measurements across RATs	37.340 (MR-DC)
38	NR (5G Radio)	5G New Radio technology	38.331 (NR RRC), 38.211 (NR PHY)

Summary

3GPP is the cornerstone of global mobile telecommunications standardization. Understanding its structure, specification numbering, and release system is essential for:

- **Engineers** implementing standards-compliant equipment
- **Researchers** building on or extending current technology
- **Product managers** planning feature roadmaps aligned with release timelines
- **Regulators** referencing normative requirements

Quick Reference Card

Question	Answer
What does 3GPP stand for?	3rd Generation Partnership Project
How many OPs?	7 (ETSI, ARIB, ATIS, CCSA, TTA, TTC, TSDSI)
How many TSGs?	3 (RAN, SA, CT)
TS vs TR?	TS = normative (shall); TR = informational (study)
Current 5G release?	Rel-18 (5G-Advanced), Rel-19 in progress
Key 5G arch spec?	TS 23.501
Key 5G NR spec?	TS 38.300 (overview), TS 38.331 (RRC)
Spec format?	xx.yyy where xx = series, yyy = spec number
Where to find?	portal.3gpp.org or 3gpp.org/specifications

End of Module 17